Mitigation of Gender and Ethnicity Bias in AI-Generated Stories through Model Explanations

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Abstract

Language models have been shown to propagate social bias through their output, particularly in the representation of gender and ethnicity. This paper investigates gender and ethnicity biases in AI-generated occupational stories. Representation biases are measured before and after applying our proposed mitigation strategy, Bias Analysis and Mitigation through Explanation (BAME), revealing improvements in demographic representation ranging from 2% to 20%. BAME leverages model-generated explanations to inform targeted prompt engineering, effectively reducing biases without modifying model parameters. By analyzing stories generated across 25 occupational groups, three large language models (Claude 3.5 Sonnet, Llama 3.1 70B Instruct, and GPT-4 Turbo), and multiple demographic dimensions, we identify persistent patterns of overrepresentation and underrepresentation linked to training data stereotypes. Our findings demonstrate that guiding models with their own internal reasoning mechanisms can significantly enhance demographic parity, thereby contributing to the development of more transparent generative AI systems.

1 Introduction

Numerous studies have shown that language models can subtly propagate social biases through generated content (Bender et al., 2021; Choubey et al., 2021). Since generative content often provides data and insights for downstream applications, its neutrality is critical even in seemingly low-stakes contexts; without such neutrality, harm can be amplified (Weidinger et al., 2021; Xing et al., 2024), perpetuating stereotypes and reinforcing societal inequities, particularly regarding gender and ethnicity (Therese Moreau and Sekara, 2024; Lee et al., 2024b; Hacker et al., 2024). In sensitive applications such as recruiting, education, and healthcare care, biased output can erode trust, lead to unfair treatment, and pose ethical and legal risks to organizations (Tilmes, 2022; Fabris et al., 2023; Pessach and Shmueli, 2020; Kheya et al., 2024; Ntoutsi et al., 2020; Chapman et al., 2022; Benatti et al., 2024; Rivera et al., 2024; Weissburg et al., 2024; Lee et al., 2024a; Gabriel et al., 2024; Sun et al., 2019).

Representation bias in generative AI occurs when underrepresented groups in training data lead to outputs that unfairly favor or disfavor certain populations, potentially reinforcing social prejudices (Shahbazi et al., 2022; Abbasi et al., 2019; Mehrabi et al., 2021; Eicher and Irgolič, 2024). Crenshaw (1989) introduced intersectionality to describe how overlapping identities like gender and ethnicity can compound discrimination (e.g, women of color). The concept now includes a wider range of social identities, showing how multiple marginalized statuses intensify bias (Buolamwini and Gebru, 2018). Furthermore, Blodgett et al. (2020) cautioned that these computational harms can cause real-world damage, including stigmatization of marginalized groups, including occupational segregation (Kirk et al., 2021).

However, addressing representation bias in generative tasks remains challenging (Kumar et al., 2024; Ferrara, 2024; Wei et al., 2025). Among the various datasets developed to evaluate representational bias in language models, BOLD and HolisticBias (Dhamala et al., 2021; Smith et al., 2022) stand out for their scope and demographic coverage. However, both lack integration of model-generated reasoning or explanations, limiting understanding of why models arrive at their outputs.

In this study, we examine bias in the generation of stories related to occupations. Occupations serve as an ideal domain for bias analysis as they are deeply intertwined with social identity, gender roles, and socioeconomic status, with well-documented historical disparities. Generated stories offer a distinct advantage as they enable rich, descriptive portrayal of characters and can encom-